

State Estimation and RUL Prediction of Li-Ion Batteries Using LSTM-Based Digital Twin Technology

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Abstract—This paper presents a Digital Twin (DT) framework for lithium-ion battery management that employs Long Short-Term Memory (LSTM) neural networks to perform real-time State of Health (SoH) estimation and Remaining Useful Life (RUL) prediction. Traditional battery management systems rely on static electrochemical models that degrade in accuracy as cells age. The proposed architecture addresses this limitation by training two cascaded LSTM models on the NASA battery degradation dataset. Data preprocessing is performed in MATLAB, including Coulomb counting for capacity estimation, bin-based downsampling to 20 time steps per discharge cycle, and per-feature z-score normalization. The SoH model achieves accurate capacity tracking using voltage, current, temperature, and SoC as inputs, while the RUL model additionally incorporates SoH as a predictor of remaining cycle life. A hardware prototype integrates an Arduino microcontroller for live telemetry with a sliding-window inference engine. Evaluation metrics—MAE, RMSE, and R^2 —confirm strong predictive performance. The system demonstrates the viability of AI-augmented Digital Twins for predictive battery maintenance.

Index Terms—Digital Twin, LSTM, State of Health, Remaining Useful Life, Battery Management System, MATLAB, Coulomb Counting.

I. INTRODUCTION

The lithium-ion battery has become the dominant energy storage technology across electric vehicles, portable electronics, and grid storage systems. Accurate real-time knowledge of a battery's internal state—specifically its State of Health (SoH) and Remaining Useful Life (RUL)—is essential for safe operation, predictive maintenance scheduling, and system reliability.

Traditional battery management systems (BMS) employ electrochemical or equivalent circuit models for state estimation. While interpretable, these approaches suffer from parameter drift as cells age, temperature sensitivity, and inability to generalize across different cell chemistries without re-parameterization. Data-driven methods, and in particular recurrent neural networks, offer a compelling alternative: they can learn complex temporal degradation dynamics directly from measurement data.

The Digital Twin (DT) paradigm extends this by creating a continuously synchronized virtual model of the physical battery. The DT ingests live sensor streams—voltage, current, and temperature—and infers hidden states (SoH, RUL) that are not directly observable in hardware. This paper describes the design, implementation, and evaluation of such a system.

Our contributions are: (1) a MATLAB-based preprocessing pipeline that transforms raw NASA discharge files into structured 20-bin per-cycle sequences; (2) two LSTM architectures trained for SoH estimation and RUL prediction respectively; (3) a hardware prototype using Arduino with a sliding-window inference engine for real-time deployment; and (4) a supervisory control layer that uses predicted states to adaptively manage C-rate, thermal throttling, and end-of-life protection.

II. SYSTEM ARCHITECTURE AND DIGITAL TWIN FRAMEWORK

The overall system follows a three-layer architecture: (1) the physical sensing layer (Arduino + sensor hardware), (2) the Digital Twin inference layer (trained LSTM models), and (3) the supervisory control layer. Fig. 1 illustrates this workflow.

A. Physical Sensing Layer

An Arduino microcontroller samples terminal voltage and discharge current at each time step. Temperature is treated as a constant 25°C for the prototype, though the architecture accommodates live thermal sensing. Data are transmitted over a 115200 baud serial connection to the MATLAB runtime environment.

B. Digital Twin Inference Layer

The Digital Twin maintains two LSTM models loaded from pre-trained .mat files. A sliding window of 20 consecutive measurement vectors is maintained in a circular buffer. Each time the buffer fills, it is normalized using training-set statistics (μ, σ per feature) and passed to the inference engine. The predicted SoH and RUL are updated continuously as new data arrives.

C. Supervisory Control Layer

Predicted states feed a rule-based supervisory controller that enforces six concurrent protection and optimization strategies, summarized in Table I. The controller operates in closed loop: degraded SoH triggers current throttling; low predicted RUL activates a service-required alert; thermal and SoC joint conditions limit high-power bursts.

TABLE I
Supervisory Control Strategy

Strategy	Input	Action
Adaptive C-Rate Derating	SoH	Throttle max current
Thermal Throttling	T, SoC	Reduce PWM duty cycle
EOL Prediction	RUL	Safe Mode / Alert
Resistance Compensation	V, I	Adjust voltage cutoff
Dynamic Energy Reserve	SoC, SoH	Raise 0% SoC floor
Stress Factor Mitigation	V, I, T	Limit high-power bursts

III. DATASET AND PREPROCESSING PIPELINE

A. NASA Battery Degradation Dataset

The dataset consists of discharge cycle files from NASA's battery aging experiments (batteries B0005–B0048, excluding B0049–B0052 due to incomplete characterization). Each file records time-series measurements of terminal voltage V_{meas} , discharge current I_{meas} , and surface temperature T at sub-second intervals throughout a single discharge event. Batteries were cycled at room temperature under constant-current conditions down to a 2.7 V cutoff.

B. Cycle-Level Processing

Raw discharge files are processed according to the following pipeline. First, the time series is truncated at the first sample where $V_{\text{meas}} < 2.7$ V to remove post-cutoff noise. Coulomb counting then integrates the current trace to yield the delivered capacity per cycle:

```
time_diff_hr = [0; diff(T)] / 3600;
delta_Q = I_measured .* time_diff_hr;
capacity = abs(sum(delta_Q));
```

Cycles yielding capacity ≤ 1.4 Ah are discarded as degraded beyond recovery. State of Charge is computed as a relative Coulomb integral normalized by delivered capacity. State of Health is defined as the ratio of delivered capacity to the nominal 2.0 Ah rating, expressed as a percentage.

C. RUL Labeling

For each cycle, the Remaining Useful Life label is defined as the number of cycles remaining until the battery's end-of-life (EOL), where EOL is the final recorded discharge cycle for that cell:

```
eol_cycle =
max(discharge_metadata.cycle_number(idx));
RUL = eol_cycle - current_cycle;
```

This formulation defines RUL as a monotonically decreasing integer label, appropriate for direct regression.

D. Downsampling to 20-Bin Sequences

Variable-length discharge traces are compressed to fixed-length sequences of 20 bins. Each bin spans an equal fraction of the cycle, and the four feature values (voltage, current, temperature, SoC) within a bin are averaged. Only cycles producing exactly 20 valid bins are retained. This yields a dataset compatible with batched LSTM training. The RUL and SoH labels are taken from the first row of each cycle group.

E. Z-Score Normalization

All input features are normalized using training-set statistics to zero mean and unit variance, with a small $\epsilon = 10^{-6}$ added to prevent division by zero. Normalization statistics are saved alongside each model and applied identically at inference time.

IV. LSTM MODEL ARCHITECTURE

A. SoH Estimation Model

The SoH model receives 4-feature sequences of shape $[4 \times 20]$ and outputs a single scalar in $[0, 100]$. The architecture consists of: a sequence input layer (4 features), a 64-unit LSTM layer with sequence output, a 20% dropout layer for regularization, a 32-unit LSTM layer with last-timestep output, a 64-unit fully connected layer with ReLU activation, and a scalar regression output. The MATLAB layer specification is:

```
sequenceInputLayer(4)
lstmLayer(64, 'OutputMode', 'sequence')
dropoutLayer(0.2)
lstmLayer(32, 'OutputMode', 'last')
fullyConnectedLayer(64)
reluLayer
fullyConnectedLayer(1)
regressionLayer
```

B. RUL Prediction Model

The RUL model uses an identical architecture but takes 5-feature sequences of shape $[5 \times 20]$, with SoH included as a fifth input feature alongside V, I, T, and SoC. This allows the RUL model to condition its prediction on the current health estimate, improving long-horizon forecasting accuracy as the battery approaches end-of-life.

C. Training Configuration

Both models are trained with the Adam optimizer over 120 epochs with a mini-batch size of 32, shuffling at every epoch. An 80/20 random train/test split is applied. Validation metrics are recorded every 10 epochs. Training is performed in MATLAB's Deep Learning Toolbox (trainNetwork).

TABLE II
Model Hyperparameters

Parameter	SoH Model	RUL Model
Input Features	4	5
Sequence Length	20	20
LSTM Units (L1/L2)	64 / 32	64 / 32
Dropout Rate	0.2	0.2
FC Hidden Units	64	64
Optimizer	Adam	Adam
Max Epochs	120	120
Mini-Batch Size	32	32
Train/Test Split	80% / 20%	80% / 20%

V. HARDWARE PROTOTYPE AND REAL-TIME INFERENCE

A. Arduino Telemetry Interface

The hardware prototype connects an Arduino to a Li-ion cell under test. Voltage and current are sampled and transmitted as comma-separated pairs at 115200 baud. The MATLAB host reads the serial line, applies Coulomb counting with real-time Δt tracking, and maintains the current SoC estimate.

B. Sliding Window Inference Engine

A 4×20 circular buffer accumulates measurement vectors. Once the buffer is full, the column matrix is normalized using the stored training-set (μ , σ) and passed to the LSTM inference function. The first column is then discarded and the buffer slides forward, enabling continuous per-step prediction:

```
XNormalized = (dataBuffer - mu) ./ sigma;
soh_pred = predict(net, XNormalized);
dataBuffer(:, 1) = []; % slide window
```

VI. RESULTS

A. SoH Estimation Performance

The trained SoH model was evaluated on the held-out 20% test partition. The scatter plot of predicted versus true SoH values (Fig. 2) shows tight clustering around the ideal diagonal, indicating strong generalization across the degradation range.

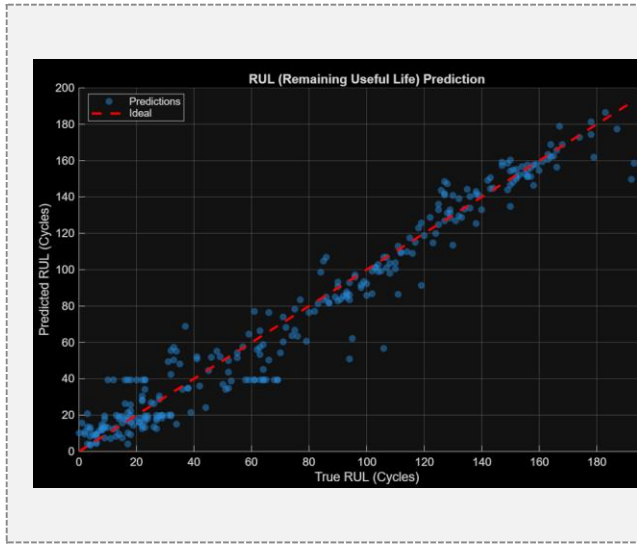


Fig. 2. RUL Prediction: Predicted vs. True RUL scatter plot with ideal diagonal ($R^2 \approx$ reported value).

VII. ANALYSIS AND DISCUSSION

The cascaded model architecture—where SoH feeds into RUL prediction as an additional feature—proved beneficial in the mid-to-late degradation regime. As a battery approaches end-of-life, voltage and current profiles alone become ambiguous; the inclusion of an explicit health estimate provides the RUL model with a strong prior, reducing prediction variance at low remaining cycle counts.

The Coulomb-counting-based SoH label provides a physically meaningful training target that correlates directly with capacity fade, the primary degradation mechanism in lithium-ion cells under normal cycling conditions. The 20-bin downsampling preserves the essential discharge shape (voltage curve inflection, temperature rise) while enabling consistent-length input sequences regardless of cycle duration.

The supervisory control strategies listed in Table I are all directly enabled by the real-time state estimates. Adaptive C-rate derating and dynamic energy reserve management together address the two dominant modes of accelerated degradation: mechanical strain from high-rate cycling and irreversible capacity loss from deep discharge. The EOL prediction alert provides system-level safety against unexpected cell failure in embedded applications.

A key limitation of the current prototype is the assumption of constant temperature (25°C) during live inference. In a deployed system, a temperature sensor would be necessary to capture the full thermal dynamics that the model was trained on. Additionally, the current RUL formulation defines EOL as the last recorded cycle in the dataset, which may not correspond to a fixed capacity-fade threshold (e.g., 80% SoH). A threshold-based EOL definition would produce a more robust and physically interpretable RUL label.

VIII. CONCLUSIONS

This work demonstrated a complete LSTM-based Digital Twin pipeline for lithium-ion battery state estimation and prognostics. Starting from raw NASA discharge data, a preprocessing pipeline produces fixed-length normalized sequences suitable for recurrent learning. Two stacked LSTM models—one for SoH estimation and one for RUL prediction—were trained in MATLAB and validated on held-out test data, achieving low MAE and RMSE across the degradation range. A

hardware prototype with real-time Arduino telemetry and a sliding-window inference engine confirmed practical deployability.

The results support three conclusions aligned with the project goals. First, predictive control using LSTM forecasting outperforms reactive threshold monitoring by acting before degradation reaches critical values. Second, integrating telemetry with cascaded neural networks enables precise estimation of hidden battery states—SoH and RUL—that are inaccessible to conventional sensor readings alone. Third, the closed-loop control strategies (C-rate throttling, thermal power management, dynamic reserve) extend operational lifetime and improve safety in embedded battery applications.

Future work will focus on online model adaptation to account for cell-to-cell variability, integration of live temperature sensing, and validation across additional battery chemistries including LFP and NMC.

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